

DATA 427 Group Project: Spark ML on the KDD Dataset

Insert ur names here

- 1 Implement two programs that apply Spark's Decision Tree algorithm and Logistic Regression algorithm to the provided KDD dataset. Run these programs 10 times on the school's Hadoop cluster, each using a different seed to split the dataset into a training and a test set.**

1.1 Describe the two programs using pseudo-code

1.1.1 Logistic Regression:

Algorithm 1 Spark Logistic Regression

- 1: Parse the input/output directories and number of runs
- 2: Initialise the Spark session and load the data from the provided input directory

Setup the pipeline as follows:

- 3: Index required columns
 - 4: One-hot encode required columns
 - 5: Assemble into feature columns
 - 6: Append parameterised logistic regression model as the final stage.
 - 7: **for** Number of runs **do**
 - 8: Set seed
 - 9: The data into train/test
 - 10: Fit the LR model on the train data
 - 11: Evaluate the models performance on the training and testing split, and record results
 - 12: **end for**
 - 13: Display results
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1.2 Describe how to install and run the two programs step by step in a Readme.txt file.

Paste readme file here I guess?

Algorithm 2 Spark Decision Tree

- 1: Parse the input/output directories and number of runs
 - 2: Initialise the Spark session and load the data from the provided input directory
Generate schema
Setup the pipeline as follows:
 - 3: Index required columns
 - 4: One-hot encode required columns
 - 5: Assemble into feature columns
 - 6: Append parameterised logistic regression model as the final stage.
 - 7: **for** Number of runs **do**
 - 8: Set seed
 - 9: The data into train/test
 - 10: Fit the Decision Tree model on the train data
 - 11: Evaluate the models performance on the training and testing split, and record results
 - 12: **end for**
 - 13: Display results
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1.3 Report the training and test results including the max, min, average accuracy and the standard deviation obtained from the 10 runs, and the running time of each program.

Table 1: Summary of model test accuracy and total running time over 10 runs.

Program	Average acc. (%)	Std. dev.	Min acc. (%)	Max acc. (%)	Total time (s)
Decision Tree	94.57	0.20	94.37	95.06	65.37
Spark Logistic Regression	94.15	0.17	93.93	94.37	129.29